

TAYLOR PENTZ

Senior Software Engineer — Angular • Node.js • TypeScript • AWS

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Professional Summary

Senior Software Engineer with 10+ years delivering **Angular 4–20** and **Node.js 8–20** solutions across **SaaS, healthcare workflow, enterprise platforms, and cloud modernization**, combining **TypeScript, NestJS/Express, AWS**, API architecture, legacy upgrades, and production reliability ownership to build secure, scalable products, stabilize complex systems, reduce delivery risk, and accelerate measurable business outcomes.

Technical Skills

- ❖ **Frontend Engineering:** **Angular 4–20, AngularJS, TypeScript 2.9–5.x**, RxJS, NgRx, Angular Material, Reactive Forms, Standalone Components, Lazy Loading, Route Guards, Micro Frontends, Webpack, Nx Monorepos, HTML5, CSS3, SCSS, Bootstrap, Tailwind CSS.
- ❖ **Backend Engineering:** **Node.js 8–20, NestJS 7–10**, Express.js, REST APIs, GraphQL, WebSockets, Fastify, TypeORM, Prisma, Sequelize, Mongoose, JWT, OAuth2/OIDC, RBAC, API Gateway patterns, background jobs, queue-based processing.
- ❖ **Cloud & DevOps:** **AWS**, ECS, EKS, Lambda, S3, CloudFront, API Gateway, RDS, ElastiCache, SQS, SNS, CloudWatch, Docker, Kubernetes, Helm, Terraform, GitHub Actions, Jenkins, GitLab CI, Nginx, Linux.
- ❖ **Databases & Caching:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis, DynamoDB, Elasticsearch/OpenSearch, database indexing, query optimization, migration scripts, data modeling, transactional consistency.
- ❖ **Testing & Quality:** Jest, Jasmine, Karma, Cypress, Playwright, Supertest, Postman/Newman, Pact basics, SonarQube, ESLint, Prettier, unit testing, integration testing, API contract testing, regression testing.
- ❖ **Monitoring & Reliability:** CloudWatch, Datadog, Sentry, Prometheus, Grafana, OpenTelemetry, ELK Stack, structured logging, distributed tracing, alert tuning, incident triage, production debugging.
- ❖ **Architecture & Practices:** Microservices, Modular Monoliths, Clean Architecture, Domain-Driven Design, CQRS, Event-Driven Architecture, CI/CD, Agile/Scrum, code review, mentoring, technical documentation, secure SDLC.

Professional Experience

SumatoSoft

Senior Software Engineer

Boston, MA (Remote) | Apr 2023 – Mar 2025

- Led modernization of customer-facing SaaS modules using **Angular 13–20, TypeScript, RxJS, NgRx**, and **Node.js 18–20**, improving frontend maintainability while preserving stable workflows during active product growth.

- Designed and delivered **NestJS** and **Express.js** backend services for account management, reporting, notifications, and billing workflows, using **PostgreSQL**, **Redis**, and **AWS ECS** to support scalable API traffic.
- Migrated legacy Angular module-based screens toward **standalone components**, route-level lazy loading, stricter TypeScript settings, and reusable UI libraries without disrupting active customer releases.
- Resolved production performance bottlenecks caused by excessive Angular change detection, duplicated API calls, and inefficient PostgreSQL joins, improving dashboard response times by **37%** across high-traffic tenant accounts.
- Implemented secure authentication and authorization flows using **JWT**, refresh-token rotation, **OAuth2/OIDC**, route guards, and role-based API policies for enterprise customer environments.
- Fixed a recurring duplicate-payment webhook issue by adding idempotency keys, transaction boundaries, retry-safe queue workers, and structured failure logging across **Node.js** services.
- Improved deployment reliability by standardizing **GitHub Actions**, Docker image builds, environment validation, migration checks, and rollback procedures across staging and production pipelines.
- Built reusable Angular form components for complex approval workflows, reducing duplicated validation logic and making product changes easier for distributed engineering teams to review and extend.
- Strengthened observability with **CloudWatch**, **Sentry**, structured logs, and API latency dashboards, allowing engineers to diagnose timeout spikes, failed jobs, and customer-impacting defects faster.
- Mentored mid-level developers on **Angular architecture**, **Node.js API design**, code review standards, testing strategy, and production-readiness practices for senior-level feature ownership.

Hexaview Technologies Inc.

Software Engineer

New York, NY | Jul 2018 – Mar 2023

- Developed enterprise web applications using **Angular 4–13**, **TypeScript**, **RxJS**, **Node.js 8–16**, **Express.js**, and **NestJS**, supporting finance, workflow automation, and internal operations platforms.
- Modernized older AngularJS and early Angular screens into component-driven Angular modules with typed services, reactive forms, shared validators, and maintainable routing structures.
- Built REST APIs for user management, document workflows, audit history, notifications, and reporting, using **Node.js**, **PostgreSQL**, **MongoDB**, and **Redis** for reliable service behavior.
- Reduced production defect leakage by **32%** by expanding Jest, Jasmine, Karma, Cypress, API integration tests, and release smoke checks around critical customer workflows.
- Solved recurring memory growth in a Node.js document-processing service by profiling heap usage, removing unbounded in-memory buffers, and moving large tasks into queue-based workers.
- Implemented migration scripts and backward-compatible API changes during database schema upgrades, preventing downtime for customers using older portal versions.
- Improved Angular performance by applying lazy-loaded feature modules, optimized RxJS subscriptions, OnPush change detection, pagination, and server-side filtering for large datasets.
- Collaborated with product managers, QA engineers, and DevOps teams to estimate features, clarify edge cases, review acceptance criteria, and deliver predictable sprint outcomes.
- Supported AWS-hosted deployments using **EC2**, **S3**, **RDS**, **CloudWatch**, Docker, Jenkins, and GitHub Actions, improving release visibility and troubleshooting discipline.

SimbirSoft Company

Software Developer

Boston, MA | Mar 2015 – Jun 2018

- Built and maintained business web applications using **AngularJS**, JavaScript, TypeScript, **Node.js**, Express.js, HTML, CSS, and SQL-backed service layers.
- Contributed to gradual UI modernization from jQuery-heavy pages into AngularJS controllers, reusable directives, services, and cleaner client-side routing patterns.
- Developed Node.js API endpoints for customer records, internal task tracking, file uploads, email notifications, and role-based access workflows across operational applications.
- Cut average page-load time by **41%** by reducing oversized payloads, adding pagination, optimizing SQL queries, compressing assets, and simplifying client-side rendering paths.
- Fixed production issues involving broken file uploads, inconsistent date formatting, session expiration errors, and missing validation on multi-step customer forms.
- Supported early migration planning from AngularJS toward newer Angular patterns by separating business logic from views and documenting reusable service boundaries.
- Improved API reliability by adding structured error handling, request validation, centralized logging, and clearer exception responses for frontend and QA teams.
- Worked closely with senior engineers to review pull requests, troubleshoot browser compatibility issues, write unit tests, and improve release documentation for client deployments.

Microsoft

Software Developer Intern

Cambridge, MA | Aug 2014 – Feb 2015

- Supported internal engineering tools using JavaScript, HTML, CSS, C#, SQL, and early single-page application patterns for team productivity and reporting workflows.
- Assisted senior developers with UI fixes, data validation improvements, and backend service updates for internal dashboards used by engineering and operations teams.
- Created small reusable JavaScript components for forms, filters, and table views, improving consistency across internal pages with limited maintenance overhead.
- Investigated defects involving incorrect search results, inconsistent browser behavior, stale cached data, and missing validation messages in internal web tools.
- Wrote SQL queries and lightweight service logic to support reporting screens, data review tasks, and operational visibility for internal stakeholders.
- Participated in code reviews, sprint planning, bug triage, and release validation while learning enterprise engineering practices in a structured software environment.
- Documented setup steps, common troubleshooting notes, and testing instructions so future interns and junior developers could contribute with fewer onboarding blockers.

Projects

Enterprise SaaS Workflow Modernization Platform

- Designed a multi-tenant workflow platform using **Angular 15–17**, **Node.js 18–20**, **NestJS**, **PostgreSQL**, **Redis**, and **AWS ECS**, replacing fragmented legacy screens with modular features, secure APIs, tenant-aware authorization, background processing, and production-grade observability.

- Implemented route-level lazy loading, reusable Angular form libraries, queue-backed notification services, idempotent webhook handling, CI/CD validation, and CloudWatch dashboards to improve release confidence and reduce operational risk.

Healthcare Operations and Scheduling Portal

- Built a healthcare operations portal using **Angular 11–13, TypeScript, Node.js, Express/NestJS, PostgreSQL, MongoDB, SQS**, Docker, and Cypress for appointment scheduling, document intake, provider workflows, and administrative reporting.
- Improved reliability by adding API contract checks, form validation rules, audit logging, role-based permissions, pagination, query optimization, and regression tests around high-risk scheduling and billing workflows.

Certifications

- **AWS Certified Solutions Architect – Associate**
- **AWS Certified Developer – Associate**
- **Microsoft Certified: Azure Developer Associate**
- **Certified Kubernetes Application Developer CKAD**
- **Professional Scrum Master I PSM I**

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science (2010 – 2014)